

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. RSA Encryption Analysis

Given $p=11$, $q=13$

- a. Compute n and $\phi(n)$
- b. Choose $e=7$ and find the private key d
- c. Encrypt the message $M=9$
- d. Decrypt the ciphertext and verify the result
- e. Analyze how changing e affects security

2. Block Cipher Mode Evaluation

Given plaintext blocks:

$P=[1010,1010,1111,1010]$, Key = $K=1100$

- a. Encrypt using ECB and CBC modes (Assume IV = 0000)
- b. Compare the ciphertext outputs
- c. Analyze which mode hides patterns better and explain why

3. DES vs 3-DES vs Blowfish Performance Study

Given:

- a. DES key size = 56 bits, time = 2 ms/block
- b. 3-DES key size = 168 bits, time = 6 ms/block
- c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

- i. Calculate total encryption time for each algorithm
- ii. Analyze trade-offs between security and performance
- iii. Recommend the best algorithm for secure real-time communication

4. Diffie-Hellman Key Exchange Analysis

Given:

- a. $p=23$, $g=5$
- b. User A private key = 6
- c. User B private key = 15
- d. Compute public keys for A and B
- e. Compute the shared secret key
- f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
- g. Suggest a secure method to prevent the attack

5. ECC vs RSA Comparison

Given:

- RSA key size = 1024 bits
- ECC key size = 160 bits (equivalent security)
- Device processing capacity = limited (IoT device)
- Analyze computational cost for both systems
- Estimate which consumes less power and memory
- Justify which cryptosystem is better for the given scenario with reasoning

RUBRICS FOR CALCULATION:

Q. No	Understanding of Concepts (25)	Experimental Design (30)	Use of Tools (20)	Analysis of Results (15)	Documentation (10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

CO2 AT1 CRYPTOGRAPHY:

1. RSA Encryption Analysis

Given: $p = 11$, $q = 13$, $e = 7$, $M = 9$

a) Compute n and $\phi(n)$

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b) Find private key (d)

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Private key: $d = 103$

c) Encrypt the message

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Ciphertext = 48

d) Decrypt

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Decrypted message = 9 (Verified)

e) Effect of changing e

- Larger $e \rightarrow$ slower encryption, similar security if chosen correctly.
- Smaller valid $e \rightarrow$ faster encryption.
- e must be coprime with $\phi(n)$.

2. Block Cipher Mode Evaluation

Given:

- Plaintext: $P = [1010, 1010, 1111, 1010]$
- Key = 1100
- IV = 0000
- Encryption: XOR with key

a) Encryption

ECB Mode

- $1010 \oplus 1100 = \mathbf{0110}$
- $1010 \oplus 1100 = \mathbf{0110}$
- $1111 \oplus 1100 = \mathbf{0011}$
- $1010 \oplus 1100 = \mathbf{0110}$

ECB Ciphertext: 0110 0110 0011 0110

CBC Mode

- $C1 = (1010 \oplus 0000) \oplus 1100 = \mathbf{0110}$
- $C2 = (1010 \oplus 0110) \oplus 1100 = \mathbf{0000}$
- $C3 = (1111 \oplus 0000) \oplus 1100 = \mathbf{0011}$
- $C4 = (1010 \oplus 0011) \oplus 1100 = \mathbf{0101}$

CBC Ciphertext: 0110 0000 0011 0101

b) Comparison

- ECB: **0110 0110 0011 0110**
- CBC: **0110 0000 0011 0101**

c) Better Mode

CBC is better because it hides repeating patterns by using the previous ciphertext block.

3. DES vs 3-DES vs Blowfish

i) Total encryption time for 1000 blocks

	Algorithm Time/block	Total Time
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DES	2 ms	2000 ms (2 s)
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3-DES	6 ms	6000 ms (6 s)
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Blowfish	1 ms	1000 ms (1 s)
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ii) Trade-off

- **DES:** Fast, weak security.

- **3-DES:** Stronger security, slowest.
- **Blowfish:** Good security and fastest.

iii) Recommendation

Blowfish is the best for secure real-time communication because it provides good security with the fastest encryption speed.

4. Diffie–Hellman Key Exchange

Given: $p = 23$, $g = 5$, A private = 6, B private = 15

a) Public keys

A =

B =

b) Shared Secret

A computes

B computes

Shared Secret Key = 2

c) Man-in-the-Middle Attack

An attacker intercepts the exchanged public keys and replaces them with their own public keys. Both users unknowingly establish separate keys with the attacker instead of each other.

d) Prevention

Use **Digital Signatures** or **Digital Certificates (PKI)** to authenticate the public keys and prevent MITM attacks.

5. ECC vs RSA Comparison

Given

- RSA = 1024 bits
- ECC = 160 bits
- IoT device (limited processing)

a) Computational Cost

- **RSA:** High computation.
- **ECC:** Low computation.

b) Power and Memory

- **RSA:** Uses more power and memory.
- **ECC:** Uses less power and memory.

c) Best Choice

ECC is better because:

- Smaller key size
- Equivalent security
- Faster computation
- Lower power consumption
- Less memory usage
- Suitable for IoT devices